INSTITUTO COLLEGE

MATH 110: General Education Math

Fall 2026

Homework 1

Name: _____

1. (*6 pts*) Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ be the universal set, and let $A = \{2, 4, 6, 8, 10\}$ and $B = \{1, 2, 3, 4, 5\}$.

   (a) Find $|A|$. **Solution:** $|A| = 5$.

   (b) Find $A \cup B$. **Solution:** $A \cup B = \{1, 2, 3, 4, 5, 6, 8, 10\}$.

   (c) Find $A \cap B$. **Solution:** $A \cap B = \{2, 4\}$.

   (d) Find A^c. **Solution:** $A^c = \{1, 3, 5, 7, 9\}$.

2. (*6 pts*) A hospital ID badge consists of a letter (A–Z), followed by a digit (0–9), followed by another letter (A–Z). Letters and digits may be repeated. How many different badges are possible?
Solution: There are 26 possibilities for the first letter, 10 possibilities for the digit, and 26 possibilities for the second letter. By the Fundamental Principle of Counting, then, there are $26 \cdot 10 \cdot 26 = 6{,}760$ possible ID badges.

3. (*6 pts*) Compute each of the following without a calculator, showing your work.

(a) $6!$
Solution: $6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 30 \cdot 12 \cdot 2 = 360 \cdot 2 = 720.$

(b) $\dfrac{8!}{6!}$
Solution: $\frac{8!}{6!} = \frac{8 \cdot 7 \cdot 6!}{6!} = 8 \cdot 7 = 56.$

(c) $\dfrac{7!}{3!\,4!}$
Solution: $\frac{7!}{3!\,4!} = \frac{7 \cdot 6 \cdot 5 \cdot 4!}{3!\,4!} = \frac{7 \cdot 6 \cdot 5}{3 \cdot 2 \cdot 1} = 7 \cdot 5 = 35.$

4. (*6 pts*) In how many different orders can 6 patient charts be stacked on a nurse's desk?
Solution: $6! = 720.$

5. (*6 pts*) A hospital needs to choose 3 nurses from a pool of 11 to form a rapid-response team. The team members have no distinct roles. How many different teams are possible?
Solution: ${}_{11}C_3 = \frac{11!}{3!\,(11-3)!} = \frac{11 \cdot 10 \cdot 9 \cdot 8!}{3!\,8!} = \frac{11 \cdot 10 \cdot 9}{3 \cdot 2 \cdot 1} = 11 \cdot 5 \cdot 3 = 55 \cdot 3 = 165.$

6. (*14 pts*) The table below shows the eight patients on a hospital ward, whether each has received a flu shot this season, and whether each has type 2 diabetes.

Patient	Flu shot	Type 2 diabetes
Ana	Yes	No
Ben	Yes	Yes
Carla	No	No
David	Yes	No
Eva	No	Yes
Frank	Yes	Yes
Gloria	No	No
Hector	Yes	No

Using first initials, let U be the set of all eight patients, let V be the set of patients who have received a flu shot, and let T be the set of patients with type 2 diabetes.

(a) (*3 pts*) Write U, V, and T by listing their elements.
Solution: $U = \{\text{A, B, C, D, E, F, G, H}\}$, $V = \{\text{A, B, D, F, H}\}$, $T = \{\text{B, E, F}\}$.

(b) (*1 pt*) Find $|U|$, $|V|$, and $|T|$.
Solution: $|U| = 8$, $|V| = 5$, $|T| = 3$.

(c) (*2 pts*) Find $V \cup T$ and $V \cap T$.
Solution: $V \cup T = \{\text{A, B, D, E, F, H}\}$, $V \cap T = \{\text{B, F}\}$.

(d) (*2 pts*) Find V^c. Describe the patients in V^c in words.
Solution: $V^c = \{\text{C, E, G}\}$, the patients who have not received a flu shot this season.

(e) (*2 pts*) Is $T \subseteq V$? Explain.
Solution: No, T is not contained in V, since $\text{E} \in T$ but $\text{E} \notin V$.

(f) (*2 pts*) Are V and T disjoint? Explain.
Solution: No, V and T are not disjoint, since $V \cap T = \{\text{B, F}\} \neq \emptyset$.

(g) (*2 pts*) Compare $|V \cup T|$ with $|V| + |T|$. Why are they different?
Solution: $|V \cup T| = |\{\text{A, B, D, E, F, H}\}| = 6$, whereas $|V| + |T| = 5 + 3 = 8$. They are different because in the sum $|V| + |T|$, the two elements of $V \cap T = \{\text{B, F}\}$ are counted twice.

7. (*14 pts*) A clinic sends each patient a confirmation code made of two letters (A–Z) followed by three digits (0–9), such as KR502.

(a) (*3 pts*) How many codes are possible if letters and digits may be repeated?
Solution: There are 26 possibilities for the first letter, 26 possibilities for the second letter, 10 possibilities for the first digit, 10 possibilities for the second digit, and 10 possibilities for the third digit. By the Fundamental Principle of Counting, then, there are $26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 = 676{,}000$ possible codes.

(b) (*3 pts*) How many codes are possible if the two letters must be different, but digits may be repeated?
Solution: Now there are only 25 possibilities for the second letter, but everything else is the same. So, there are $26 \cdot 25 \cdot 10 \cdot 10 \cdot 10 = 650{,}000$ possible codes.

(c) (*4 pts*) How many codes are possible if no letter and no digit may be repeated?
Solution: There are 26 possibilities for the first letter, 25 possibilities for the second letter, 10 possibilities for the first digit, 9 possibilities for the second digit, and 8 possibilities for the third digit. Therefore, there are $26 \cdot 25 \cdot 10 \cdot 9 \cdot 8 = 468{,}000$ possible codes.

(d) (*4 pts*) In part (c), the digits available for the second digit depend on which digit was chosen first. Explain why the Fundamental Principle of Counting can still be used.
Solution: Even though the digits available for the second digit depend on which digit was chosen first, the *number* of digits available does not. There are 9 available digits regardless of which digit was chosen first.

8. (*14 pts*)

(a) (*4 pts*) Compute $1!$, $2!$, $3!$, $4!$, $5!$, and $6!$.
Solution: $1! = 1 \cdot 0! = 1 \cdot 1 = 1$, $2! = 2 \cdot 1! = 2 \cdot 1 = 2$, $3! = 3 \cdot 2! = 3 \cdot 2 = 6$, $4! = 4 \cdot 3! = 4 \cdot 6 = 24$, $5! = 5 \cdot 4! = 5 \cdot 24 = 120$, $6! = 6 \cdot 5! = 6 \cdot 120 = 720$.

(b) (*4 pts*) Simplify $\dfrac{10!}{8!}$ without computing $10!$ or $8!$. Show your work.
Solution: $\frac{10!}{8!} = \frac{10 \cdot 9 \cdot 8!}{8!} = 10 \cdot 9 = 90$.

(c) (*3 pts*) Simplify $\dfrac{100!}{98!}$.
Solution: $\frac{100!}{98!} = \frac{100 \cdot 99 \cdot 98!}{98!} = 100 \cdot 99 = 9{,}900$.

(d) (*3 pts*) A student claims that $\dfrac{6!}{3!} = 2!$, since $\dfrac{6}{3} = 2$. Show that the student is wrong by computing both sides.
Solution: $\frac{6!}{3!} = \frac{6 \cdot 5 \cdot 4 \cdot 3!}{3!} = 6 \cdot 5 \cdot 4 = 30 \cdot 4 = 120$, which is not equal to $2! = 2$.

9. (*14 pts*) A pharmacy has 12 different medications on a shelf, but there is only room to display 5 of them in the front display case, arranged in a row from left to right.

(a) (*3 pts*) How many choices are there for the first (leftmost) spot in the display? For the second spot? For the third, fourth, and fifth spots?
Solution: There are 12 choices for the first spot, 11 for the second, 10 for the third, 9 for the fourth, and 8 for the fifth.

(b) (*2 pts*) Use the Fundamental Principle of Counting and your answer to part (a) to find the number of possible displays.
Solution: There are $12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 = 95{,}040$ possible displays.

(c) (*3 pts*) Write your answer to part (b) as a permutation number ${}_nP_k$, and check it using the formula ${}_nP_k = \dfrac{n!}{(n-k)!}$.
Solution: ${}_{12}P_5 = \frac{12!}{(12-5)!} = \frac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7!}{7!} = 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 = 95{,}040$, which agrees with part (b).

(d) (*3 pts*) Suppose instead the display case had room for all 12 medications. How many displays would be possible? (You may leave your answer as a factorial.)
Solution: There are $12!$ possible displays.

(e) (*3 pts*) Write your answer to part (d) as a permutation number ${}_nP_k$. What does the formula ${}_nP_k = \dfrac{n!}{(n-k)!}$ give in this case, and why does it agree with part (d)?
Solution: ${}_{12}P_{12} = \frac{12!}{(12-12)!} = \frac{12!}{0!} = 12!$. This works because $0! = 1$.

10. (*14 pts*) A clinic has 7 nurses, and 3 of them will be chosen to attend a training session.

(a) (*3 pts*) Suppose the 3 nurses chosen will present at the training first, second, and third. In how many ways can they be chosen? Is this a permutation or a combination problem?
Solution: This is a permutation problem, because order matters. The nurses can be chosen in ${}_7P_3 = \frac{7!}{(7-3)!} = \frac{7\cdot6\cdot5\cdot4!}{4!} = 7\cdot6\cdot5 = 7\cdot30 = 210$ ways.

(b) (*3 pts*) Suppose instead the 3 nurses chosen will simply attend, with no distinct roles. In how many ways can they be chosen? Is this a permutation or a combination problem?
Solution: This is a combination problem, since order does not matter. The nurses can be chosen in ${}_7C_3 = \frac{7!}{3!\,(7-3)!} = \frac{7\cdot6\cdot5\cdot4!}{3!\,4!} = \frac{7\cdot6\cdot5}{3\cdot2\cdot1} = 7\cdot5 = 35$ ways.

(c) (*3 pts*) Your answer to part (a) should be 6 times your answer to part (b). Explain why.
Solution: Indeed, $210 = 6\cdot35$. This is because ${}_7P_3 = 3!\cdot{}_7C_3$. There are ${}_7C_3$ ways to choose the three nurses, and if order matters then there are ${}_3P_3 = 3! = 6$ ways to permute them.

(d) (*3 pts*) Compute ${}_7C_4$ and compare it with your answer to part (b). Explain in words why choosing 3 nurses to attend the training is the same as choosing 4 nurses to stay at the clinic.
Solution: ${}_7C_4 = \frac{7!}{4!\,(7-4)!} = \frac{7!}{4!\,3!} = \frac{7\cdot6\cdot5\cdot4!}{4!\,3!} = \frac{7\cdot6\cdot5}{3\cdot2\cdot1} = 7\cdot5 = 35$. This is the same as ${}_7C_3$ because the factorials that appear in the denominator are the same. In words: each choice of 3 nurses to attend the training leaves exactly one group of 4 nurses to stay at the clinic, and each choice of 4 nurses to stay leaves exactly one group of 3 to attend, so the two counts must be equal. In general, ${}_nC_k = {}_nC_{n-k}$.

(e) (*2 pts*) A coin is tossed 5 times. How many different sequences of heads and tails have exactly 2 heads?
Solution: There are ${}_5C_2 = \frac{5!}{2!\,(5-2)!} = \frac{5\cdot4\cdot3!}{2!\,3!} = \frac{5\cdot4}{2} = 5\cdot2 = 10$ ways to choose the two slots in the sequence to place the heads.